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Report of

END-OF-STUDIES PROJECT

In partial fulfillment for:

Applied Bachelor's Degree in Information Systems Development

Development of an Internal Platform for Managing Requests, Projects, and Tasks at ArabSoft

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Dedication

To all those whose presence illuminated every step of this journey…

To my father, Adel,

You taught me that hard work and perseverance
are the keys to every success.
Your constant support and trust in me
have been my anchor throughout this journey.
This work is the result of your precious guidance.

To my mother, Amira,

An endless source of love and encouragement,
your caring presence has always carried me.
Every effort I made contains a part of you,
because you have been my greatest strength.

To my sister, Sarra,

For your joy, your support, and your generosity.
With one smile, you always gave me courage
in the most difficult moments.
Your presence is a precious gift.

To all my friends and colleagues, and to everyone who contributed,
directly or indirectly, to the completion of this work.

Abderrahmen Ben Hattab

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General Introduction

Digital transformation has become a major strategic driver for companies seeking to improve organizational efficiency and ensure smooth management of internal processes. The centralization of administrative requests, project tracking, and internal communication represents significant challenges when operations are handled manually or in a fragmented manner. The growing complexity of internal workflows, together with the need for effective coordination between departments, makes the adoption of modern and structured digital tools essential.

In this context, my work focuses on the **development of an internal platform for managing requests, projects, and tasks at ArabSoft**. The objective is to design a practical and reliable solution that centralizes administrative requests, monitors project progress, and improves communication between employees, team leaders, and HR managers. By providing a global and real-time view of organizational activities, this platform aims to strengthen collaboration, improve process efficiency, and support better operational decision-making.

The ambition of this project is to provide a tool that addresses current organizational needs while remaining adaptable to future evolution. Whether for handling internal requests, tracking project progress, assigning tasks, or improving information flow between stakeholders, the platform is designed as a complete and intuitive solution that integrates smoothly into existing work practices.

This report presents the different phases of project realization. From requirements analysis to deployment, including design, development, and testing phases, I explain how this platform was built to meet user expectations and deliver concrete added value to the company. Particular attention is given to the methodological approach, technical choices, and implementation strategy adopted throughout the project.

Through this work, I aim to contribute to the modernization of internal management practices and to the continuous improvement of organizational performance. This project represents an opportunity to rethink how administrative and project activities are managed by making them more structured, collaborative, and efficient.

Chapter 1

Project Context Presentation

I. Host organization presentation

1. General presentation of ArabSoft

ArabSoft is a company specialized in software engineering and digital solutions for organizations. It operates in several areas such as business process management, enterprise application development, systems integration, and digital transformation support. Through its experience in information systems, the company helps institutions modernize operations and improve service quality.

ArabSoft combines **technological expertise** and **business understanding** to deliver reliable and scalable solutions adapted to client needs. With a structured project approach and multidisciplinary teams, the company supports organizations in optimizing internal workflows, improving traceability, and increasing overall operational efficiency.

Figure 1: ArabSoft logo

2. ArabSoft in Tunisia

In Tunisia, **ArabSoft** supports both public and private organizations in the digitalization of their internal processes. Its teams work on the design, development, deployment, and maintenance of business-oriented platforms, with a focus on security, scalability, and long-term

maintainability. Key missions include process automation, workflow management, data centralization, and decision support through structured reporting.

II. Project presentation

1. Current-state study

At present, internal process management at ArabSoft relies on a combination of office tools and partially connected workflows. Daily practices include Excel files, email exchanges, messaging channels, and manual follow-up sheets to handle administrative requests and monitor project execution.

Although these tools support day-to-day operations, information remains distributed across multiple sources and teams. This fragmentation limits process visibility, slows coordination, and increases the risk of inconsistencies between request status, project progress, and assigned tasks.

The current operational flow mainly covers:

- **Administrative request handling**, to receive, validate, and process internal service requests.
- **Project and task follow-up**, to plan activities, assign responsibilities, and track execution progress.
- **Operational coordination**, to synchronize information between employees, team leaders, and HR managers and ensure timely communication.

2. Current-state critique

Despite the use of multiple tools, several difficulties remain and reduce operational efficiency. Main issues include:

- **Data fragmentation:** Request, project, and task information is spread across multiple files and channels, which makes a coherent global view difficult.
- **Delayed follow-up:** Manual tracking slows the handling of requests and makes it harder to monitor priorities, deadlines, and responsibilities.

- **Inefficient coordination:** Employees, team leaders, and HR managers do not share a single operational workspace, which increases duplication and communication gaps.
- **Limited access to updated information:** Stakeholders do not always have real-time access to the latest status, which reduces responsiveness.

These limitations show that the current tools and processes no longer fully meet ArabSoft's needs for traceability, responsiveness, and coordination in the management of internal requests, projects, and tasks.

3. Proposed solution

To address these challenges, I propose the development of an **internal platform for managing requests, projects, and tasks at ArabSoft**. The platform is designed to centralize operational data, structure workflow execution, and improve coordination across departments through a practical and intuitive environment. Key features include:

- **Data centralization:** All requests, projects, and tasks are stored in a unified database accessible in real time.
- **Workflow management:** Structured request lifecycle with clear statuses, validation steps, and processing history.
- **Project and task tracking:** Planning, assignment, prioritization, and progress monitoring through dedicated views.
- **Role-based access:** Secure access control according to user profiles and organizational responsibilities for employees, team leaders, and HR managers.
- **Notifications and integrations:** Alerts and automated exchanges with external services when needed to support authentication, messaging, and reporting.
- **Dashboards and reporting:** Operational indicators for monitoring performance, identifying bottlenecks, and supporting decision-making.

This solution provides a global real-time view of internal activities and improves efficiency, responsiveness, and traceability across the organization.

Modular, secure, and scalable, the platform is designed to adapt to ArabSoft's evolving needs while enabling progressive adoption and continuous process improvement.

III. Adopted methodologies and formalism

1. Modeling and design methodology

For project modeling and design, I chose **UML (Unified Modeling Language)**. UML is a standardized visual language used to represent both system structure and behavior.

Using UML in this project allows me to:

- **Visualize requirements:** Use-case diagrams capture expected functionalities and actor-system interactions.
- **Structure the system:** Class and sequence diagrams define module architecture and data/interaction flows.
- **Facilitate communication:** UML provides a common language for developers, project managers, and clients.

UML was selected for its flexibility, standardization, and ability to fit complex project needs.

Figure 2: UML logo

2. Working methodology

For project management, I adopted the **Agile** methodology, specifically the **Scrum** framework. This iterative and incremental approach supports requirement changes and continuous user feedback.

2.1. Methodological choice (Scrum)

Scrum was chosen for several reasons:

- **Flexibility:** Scrum adapts quickly to change, which is critical when requirements evolve.
- **Transparency:** Regular meetings such as daily stand-ups and sprint reviews ensure clear communication.
- **Continuous delivery:** Sprint cycles provide continuous validation of implemented features.

2.2. Why Scrum?

Scrum is well suited to this project because it supports multidisciplinary teams and collaboration. It also helps to:

- **Maximize value:** Focus on high-priority features aligned with project objectives.
- **Improve quality:** Frequent feedback and regular testing enable rapid issue detection and correction.

2.3. Scrum team roles

In this project, Scrum roles are distributed as follows:

- **Product Owner:** Defines priorities and manages the backlog to ensure delivered features meet user needs. In this project, the Product Owner is **Mr. Mouadh Zidi**, my company supervisor.
- **Scrum Master:** Ensures Scrum principles are followed and helps remove obstacles. In this project, the Scrum Master is **Ms. Anissa CHALOUAH**, my ISET supervisor.
- **Development Team:** Responsible for design, development, and testing of platform features. In this project, the development team is composed of **Abderrahmen Ben Hattab** only.

This role distribution enables efficient collaboration and clear responsibilities, contributing directly to project success.

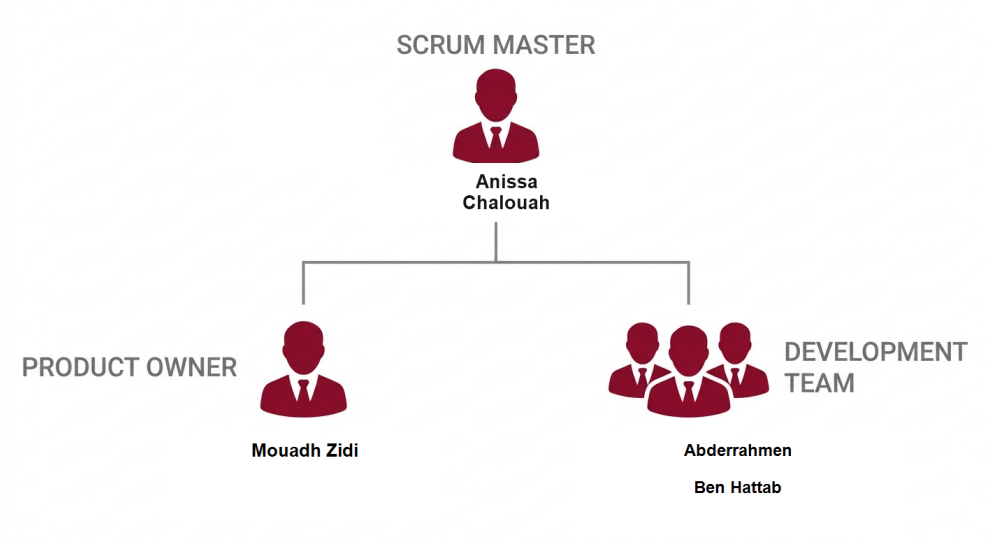


Figure 3: Scrum team roles

IV. Development environment

To complete this project, an appropriate development environment was set up. It includes hardware resources, software tools, and modern technologies to ensure effective design, development, and deployment.

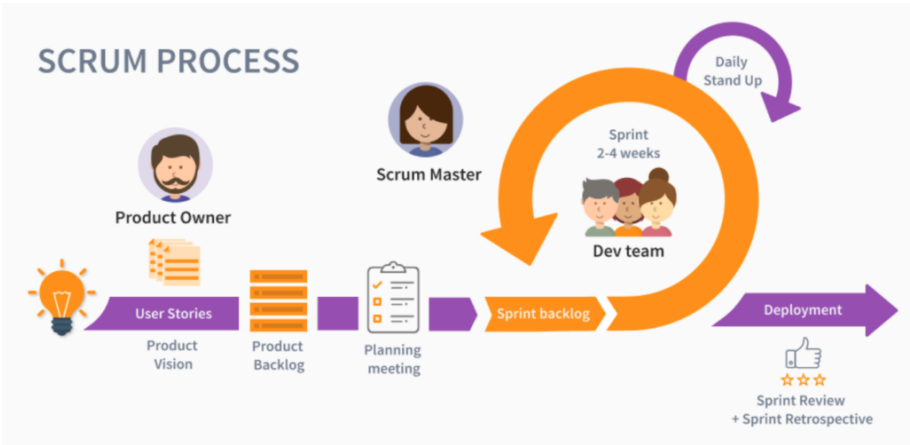
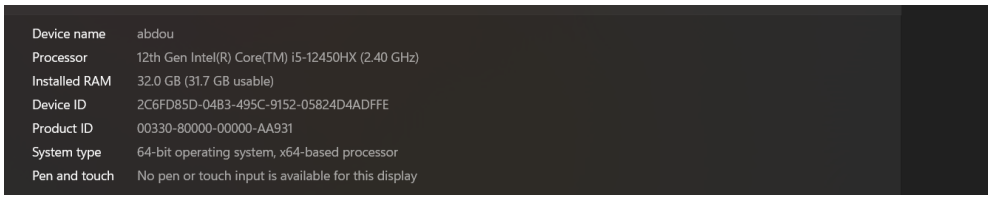


Figure 4: SCRUM method

1. Hardware environment

This project was developed on a laptop with the following specifications:



Device name	abdou
Processor	12th Gen Intel(R) Core(TM) i5-12450HX (2.40 GHz)
Installed RAM	32.0 GB (31.7 GB usable)
Device ID	2C6FD85D-04B3-495C-9152-05824D4ADFFE
Product ID	00330-80000-00000-AA931
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

Figure 5: PC specifications used for development

2. Software environment

For developing the internal request, project, and task management platform, several software tools were used. Each tool was selected for its specific features and its fit with project requirements.

- **Visual Studio Code :**

Main code editor used for **frontend** development with **React** and **Tailwind CSS**. Its lightweight design and extension ecosystem provide a productive environment for UI implementation.

- **IntelliJ IDEA :**

Integrated development environment used for **backend** development with **Spring Boot** and **PostgreSQL**. IntelliJ IDEA provides advanced autocompletion, debugging, refactoring, and native integration with Maven and Spring.

- **Postman :**

Used to test and validate REST APIs developed with Spring Boot through an intuitive interface and strong debugging and documentation features.

- **Draw.io :**

Used to design UML diagrams such as use-case, class, and sequence diagrams, as well as functional mockups, thanks to its ease of use and free access.

- **Git and GitHub :**

Used for version control and source-code management, with reliable workflows and easy integration in Visual Studio Code and IntelliJ IDEA.

- **Jira :**

Used for task management and Scrum sprint tracking. It supports user-story planning, sprint monitoring, and progress reporting.

- **LaTeX :**

LaTeX was selected for report writing to ensure a professional layout, accurate references, and typographic consistency throughout the document.

- **Figma :**

Used to create wireframes for the user interface. It enables quick visual prototyping and ergonomic validation before development.

3. Technologies used

The following technologies were chosen for their fit with project requirements and their ability to deliver a modern, high-performance solution.

- **React :**

Main frontend library used in the View layer of the platform interface. React provides reusable components, efficient rendering, and flexible state management for scalable user interfaces.

- **Tailwind CSS :**

Utility-first CSS framework used for interface styling and layout. Tailwind CSS enables fast, consistent, and responsive UI development while supporting design customization.

- **Spring Boot :**

Java framework used for **backend** development based on a **layered architecture**. Spring Boot simplifies secure and high-performance REST API development through auto-configuration and a rich ecosystem.

- **PostgreSQL :**

Relational database management system used for platform data storage and management. PostgreSQL was selected for reliability, performance, and strong support for complex queries and data integrity.

- **Keycloak :**

Identity and access management solution used to handle secure authentication, role-based authorization, and token-based security for the platform.

- **Docker :**

Containerization platform used to package and run application services consistently across development and deployment environments.

- **Kafka :**

Distributed event-streaming platform used for asynchronous communication, especially for notifications and background event processing in the system.

V. Application architecture

1. Global architecture

The application follows a web-based architecture composed of two complementary parts: a **React frontend structured with MVVM** and a **layered** backend. This organization fits an internal platform used by employees, team leaders, and HR managers to manage requests, projects, tasks, notifications, and dashboards. It ensures clear separation of responsibilities, smooth integration through REST APIs, and maintainable system evolution. External services such as Keycloak, Kafka, JasperReports, and email notifications are integrated when needed to support authentication, event handling, reporting, and communication.

2. Frontend part: MVVM architecture with React

The frontend part is implemented with **React** and **Tailwind CSS** and organized according to the **MVVM** pattern. This choice is appropriate for a role-aware internal platform where employees, team leaders, and HR managers access different views over the same operational data. React provides the View layer, view models coordinate presentation logic and state, and service classes handle communication with the Spring Boot REST API. This organization ensures a clear separation between interface rendering, presentation behavior, and service communication, which improves maintainability and functional consistency.

- **View layer:** Contains React pages and reusable UI components used to display request forms, task boards, project dashboards, notification panels, and role-specific

interfaces.

- **ViewModel layer:** Handles presentation logic, state coordination, form behavior, interaction flows, and the transformation of backend data into structures adapted to the interface.
- **Model / Service layer:** Groups frontend data models, DTO-shaped structures when needed, and service classes or functions responsible for calling the backend API and managing request and response data.

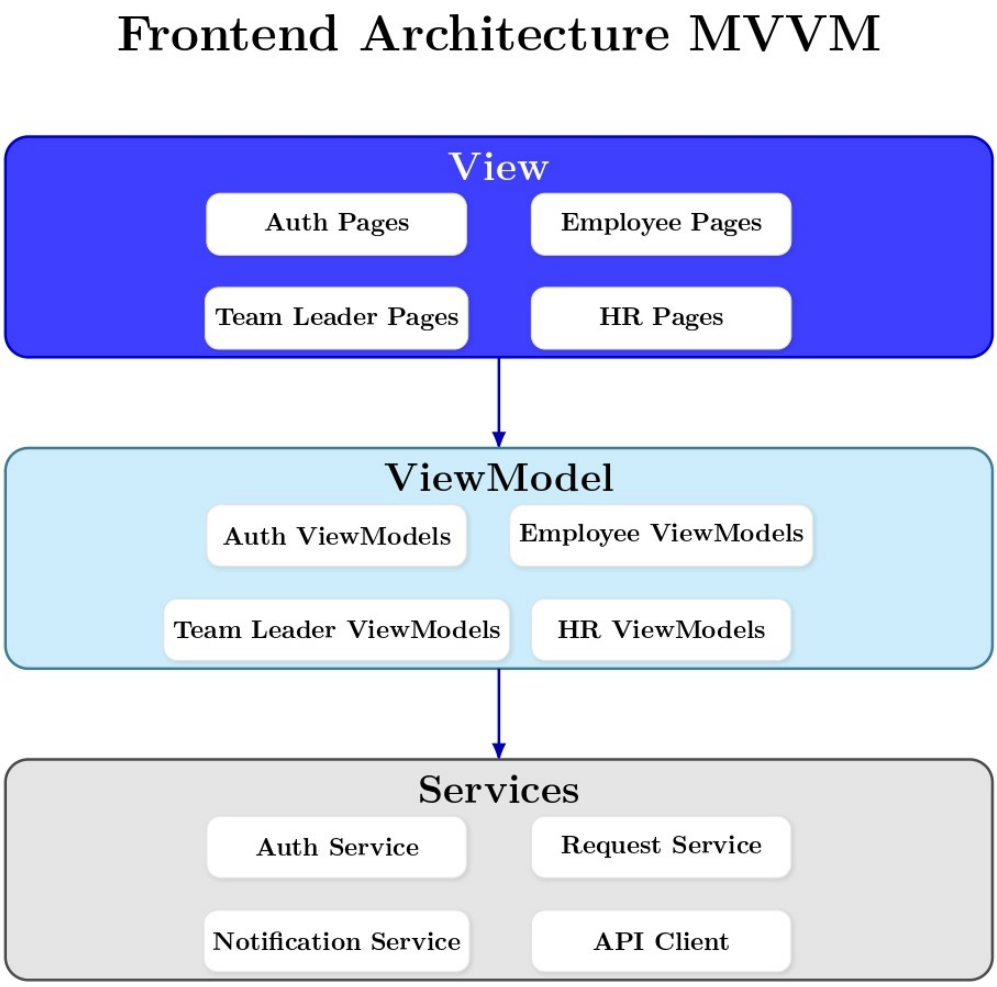


Figure 6: MVVM frontend architecture with React

3. Backend part: layered architecture with Spring Boot

The backend is based on a **layered architecture**, implemented with **Spring Boot** and **PostgreSQL**. This architecture structures the backend into clear functional layers, improving maintainability, testability, and scalability.

- **Controller layer:** Spring MVC REST controllers expose API endpoints and handle HTTP requests from the React frontend.
- **Service layer:** Contains business logic, validation rules, and orchestration of application use cases.
- **Repository layer:** Spring Data JPA repositories handle persistence and data access in **PostgreSQL**.

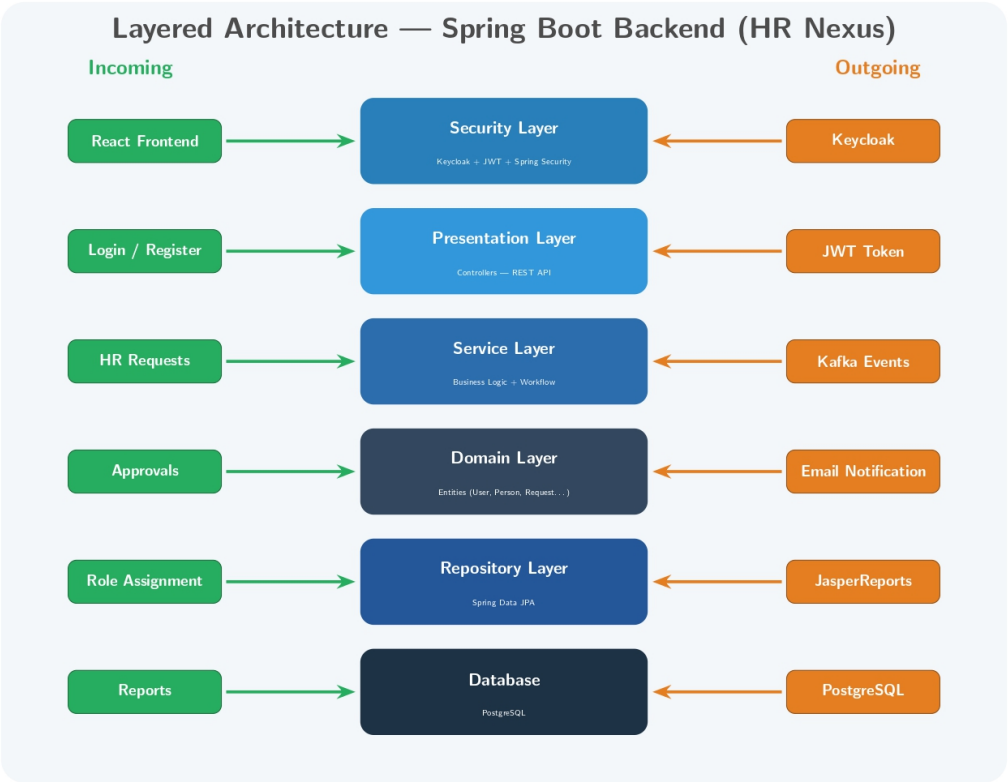


Figure 7: Layered architecture with Spring Boot

4. Why these architectures?

These architectures were selected for several reasons:

- **Separation of concerns:** MVVM separates the interface, presentation logic, and service access in the frontend, while the backend keeps controllers, services, and repositories clearly isolated.
- **Maintainability and testability:** View models can be tested and evolved independently from React pages, and backend service logic remains independent from persistence concerns.
- **Reusability and consistency:** A structured frontend makes it easier to reuse UI components and shared presentation logic across requests, projects, tasks, dashboards, and notifications.
- **Scalability and integration:** React communicates efficiently with the Spring Boot REST API, which supports progressive extension of the platform without compromising organization.

5. Global architecture illustration

The figure below presents the global application architecture in a cleaner and more report-ready form. It shows the React frontend structured with MVVM, the Spring Boot backend organized in layers, PostgreSQL as the persistent storage, and the external services integrated to support authentication, asynchronous communication, reporting, and notifications. The diagram also highlights the role-based usage of the platform by employees, team leaders, and HR managers.

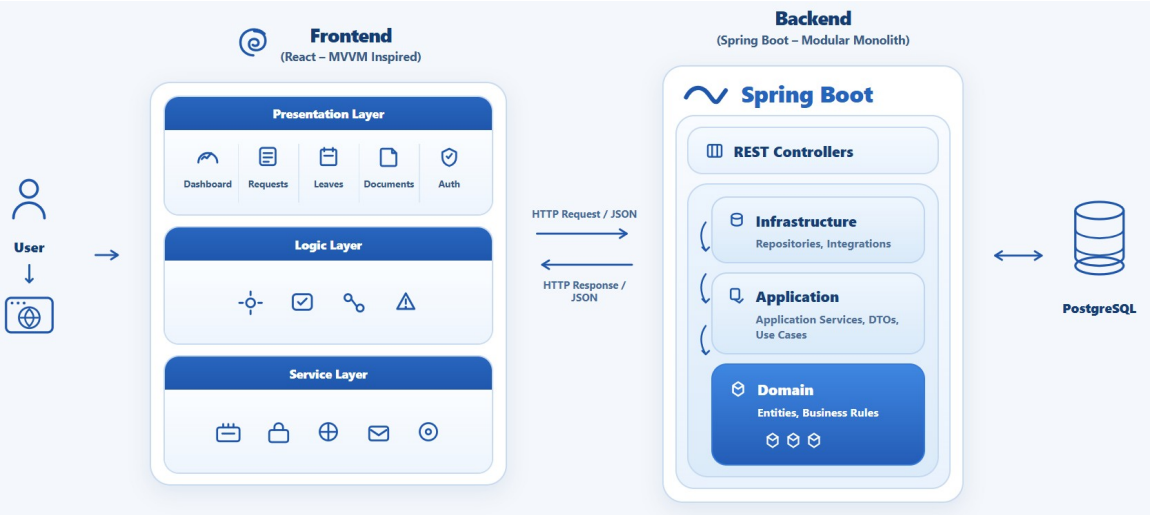


Figure 8: Global application architecture

Conclusion

This chapter presented the general project context: the host organization ArabSoft, the current internal management challenges related to requests, projects, and tasks, the proposed solution, and the adopted methodologies and development environment. These elements provide the foundation for the rest of the work. The next chapter focuses on product backlog planning and functional requirement identification.

Chapter 2

Product Backlog Planning

Introduction

This chapter defines the product backlog planning for the proposed internal platform at ArabSoft. It identifies the actors, structures the functional and non-functional requirements, and formalizes user stories and backlog priorities that will guide sprint planning and implementation.

I. Actor Identification

1. Employee

The employee submits administrative requests, tracks processing status, accesses assigned tasks, and receives notifications related to approvals, rejections, and project activities.

2. Team Leader

The team leader supervises team activities, validates requests requiring managerial approval, assigns and follows tasks, and monitors project execution indicators.

3. HR Manager

The HR manager manages employee records and roles, performs final validation of administrative requests, and monitors compliance through dashboards and reports.

4. New User

A new user is registered in the platform but remains in a restricted state until HR validation and role assignment are completed.

II. Functional Requirements

1. Authentication and access management

The system shall support registration, secure authentication, logout, account activation by HR, and role-based access control according to user profile.

2. Administrative request management

The platform shall manage submission, validation, rejection, and history tracking for leave requests, document requests, loan requests, and authorization requests.

3. Team and employee management

The system shall provide profile management, team assignment, leader designation, and maintenance of employee administrative information.

4. Project and task management

The platform shall support project creation, task assignment, workload organization, progress tracking, and status visibility for all authorized stakeholders.

5. Notification and reporting

The system shall send operational notifications and provide reports and dashboards to support monitoring, traceability, and decision-making.

6. AI assistance and decision support

The platform shall provide AI-assisted support to help employees prepare administrative requests, assist team leaders in reviewing requests and workload constraints, and support HR managers with recommendation-oriented decision assistance while preserving human validation, traceability, and auditability.

III. Non-Functional Requirements

1. Security

The application must guarantee secure authentication, protection of sensitive data, strict role-based permissions, and complete audit traceability.

2. Performance

The platform must maintain acceptable response times under expected internal usage and support concurrent request processing without significant delays.

3. Usability

Interfaces must be clear and intuitive to reduce training effort and facilitate adoption by employees, team leaders, and HR managers.

4. Maintainability

The system architecture must allow efficient correction, enhancement, and extension through modular and well-documented components.

5. Scalability

The platform should support future growth in users, workflows, and modules without requiring major redesign.

IV. User Stories

1. Employee User Stories

Employee User Stories

As an employee, I want to sign in securely so that I can access the platform according to my role.	As an employee, I want to sign out securely so that I can protect my account after use.	As an employee, I want to manage my profile information so that my personal data stays accurate.
As an employee, I want to change my password securely so that I can keep my account protected.	As an employee, I want to access my personal dashboard so that I can quickly view the most relevant information.	As an employee, I want to submit and cancel a leave request before decision so that my leave workflow stays accurate.
As an employee, I want to submit a document request so that I can obtain official HR documents.	As an employee, I want to submit a loan request so that I can ask for financial support through the platform.	As an employee, I want to submit an authorization request so that I can formalize exceptional permissions.

As an employee, I want to track request status and history so that I can follow leave and administrative workflows.

As an employee, I want to view my assigned tasks and project calendar so that I can organize my work efficiently.

As an employee, I want to filter tasks by project so that I can focus on the work context concerned.

As an employee, I want to receive notifications and emails about role, request, leave, and task updates so that I stay informed.

2. Team Leader User Stories

Team Leader User Stories

As a team leader, I want to sign in securely so that I can access the platform according to my role.

As a team leader, I want to sign out securely so that I can protect my account after use.

As a team leader, I want to manage my profile information so that my personal data stays accurate.

As a team leader, I want to change my password securely so that I can keep my account protected.

As a team leader, I want to access my dashboard so that I can quickly monitor team activity.

As a team leader, I want to review team leave requests so that I can process them according to team constraints.

As a team leader, I want to approve or reject leave requests with a reason so that decisions remain clear and traceable.

As a team leader, I want to monitor team workload so that I can balance responsibilities effectively.

As a team leader, I want to assign tasks to team members so that project work is distributed properly.

As a team leader, I want to follow task progress and receive completion notifications so that I can monitor delivery.

As a team leader, I want to view the team leave calendar within my scope so that I can anticipate availability issues.

As a team leader, I want to use employee-side leave and request actions for my own needs so that personal workflows remain consistent.

3. HR Manager User Stories

HR Manager User Stories

As an HR manager, I want to sign in securely so that I can access the platform according to my role.

As an HR manager, I want to sign out securely so that I can protect my account after use.

As an HR manager, I want to manage my profile information so that my personal data stays accurate.

As an HR manager, I want to change my password securely so that I can keep my account protected.

As an HR manager, I want to access my dashboard so that I can monitor operational activity.

As an HR manager, I want to assign roles and trigger user notification so that access changes are communicated clearly.

As an HR manager, I want to activate or deactivate accounts so that access remains controlled.

As an HR manager, I want to manage employee administrative information so that records stay complete and accurate.

As an HR manager, I want to make final decisions on leave and administrative requests so that workflows are completed properly.

As an HR manager, I want to manage teams and team leaders so that the organizational structure remains consistent.

As an HR manager, I want broad leave calendar visibility and filters so that I can supervise teams, employees, dates, and statuses.

As an HR manager, I want to consult login history and available audit traces so that system activity remains supervised.

As an HR manager, I want to upload official request files and notify users so that final HR outputs are communicated.

As an HR manager, I want to trace request history and cancellations before decision so that workflow follow-up remains clear.

V. Product Backlog

The product backlog is organized into six functional sprints. Each sprint groups related features according to business value and implementation logic, from platform access and request workflows to HR administration, project tracking, and AI-assisted support.

Sprint	Num	User Story	Description	Complexity
Sprint 1 Access	**US 1**	**Registration**	As a new user, I want to create an account so that I can request platform access.	Medium
	US 2	**Secure login**	As a user, I want to sign in securely so that I can access the platform according to my role.	Low
	US 3	**Logout**	As a user, I want to sign out securely so that I can protect my account after use.	Low
	US 4	**Password reset**	As a user, I want to reset my password securely so that I can recover access.	Medium
	US 5	**Profile update**	As a user, I want to update my profile so that my personal data stays accurate.	Low
	US 6	**NEW_USER restriction**	As an HR manager, I want new users to remain restricted until HR validation.	High
	US 7	**Role assignment**	As an HR manager, I want to assign roles so that each user receives the correct access rights.	Medium
	US 8	**Role assignment notification and email**	As a user, I want to receive a notification and email when my role is assigned.	Low
	US 9	**Account activation**	As an HR manager, I want to activate approved accounts so that validated users can access the platform.	Medium
	US 10	**Account deactivation**	As an HR manager, I want to deactivate accounts so that access can be revoked when needed.	Medium
	US 11	**Login history**	As an HR manager, I want to consult login history so that authentication activity can be traced.	Medium

Table 1: Product Backlog

Sprint	Num	User Story	Description	Complexity
Sprint 2 Leave Management	**US 12**	**Leave submission**	As an employee, I want to submit a leave request so that I can ask for time off through a structured workflow.	Medium
	US 13	**Leave tracking**	As an employee, I want to track my leave request status so that I can follow its progress.	Low
	US 14	**Leave history**	As an employee, I want to view my leave history so that I can review my previous leave requests in one place.	Low
	US 15	**Team review**	As a team leader, I want to review team leave requests before making a decision.	Medium
	US 16	**Team approval**	As a team leader, I want to approve leave requests so that valid requests can move forward in the workflow.	Medium
	US 17	**Team rejection with reason**	As a team leader, I want to reject leave requests with a reason so that the decision remains traceable.	Medium
	US 18	**HR final decision**	As an HR manager, I want to make the final decision on leave requests so that the workflow is completed properly.	High
	US 19	**Leave cancellation before decision**	As an employee, I want to cancel a leave request before a decision is taken so that I can withdraw an unnecessary request.	Low
	US 20	**Leave decision notifications and emails**	As an employee, I want notifications and emails about leave approvals or rejections.	Low
	US 21	**Team leader personal leave actions**	As a team leader, I want to submit and follow my own leave requests as an employee so that my personal requests follow the same workflow.	Low

Table 1: Product Backlog (continued)

Sprint	Num	User Story	Description	Complexity
Sprint 3 Administrative Requests	**US 22**	**Document submission**	As an employee, I want to submit a document request so that I can obtain official HR documents through the platform.	Medium
	US 23	**Loan submission**	As an employee, I want to submit a loan request so that I can ask for financial support through a structured workflow.	Medium
	US 24	**Authorization submission**	As an employee, I want to submit an authorization request so that I can formalize exceptional permissions.	Medium
	US 25	**Request tracking**	As an employee, I want to track the status of my administrative requests so that I can follow their progress.	Low
	US 26	**Request history**	As an employee, I want to view my administrative request history so that I can consult previous submissions.	Low
	US 27	**HR final decision**	As an HR manager, I want to make the final decision on administrative requests so that each request receives an official outcome.	High
	US 28	**Official file upload**	As an HR manager, I want to upload finalized files so that approved requests include official documents.	Medium
	US 29	**Request status notifications and emails**	As a user, I want notifications and emails about request status changes and decisions.	Low
	US 30	**Request cancellation before decision**	As an employee, I want to cancel an administrative request before a decision is taken so that I can withdraw it when it is no longer needed.	Low
	US 31	**Team leader personal request actions**	As a team leader, I want to submit and follow my own administrative requests as an employee so that my personal requests follow the same logic.	Low

Table 1: Product Backlog (continued)

Sprint	Num	User Story	Description	Complexity
Sprint 4 HR Management & Calendar	**US 32**	**Employee records**	As an HR manager, I want to manage employee information so that records stay accurate.	Medium
	US 33	**Team management**	As an HR manager, I want to manage teams so that the organizational structure remains consistent.	Medium
	US 34	**Team leader assignment**	As an HR manager, I want to assign team leaders so that each team has the correct supervisor.	Medium
	US 35	**Role management**	As an HR manager, I want to manage roles so that access rights match responsibilities.	Medium
	US 36	**HR dashboard**	As an HR manager, I want to access an HR dashboard so that I can monitor operational activity at a glance.	Medium
	US 37	**Login history and audit traces**	As an HR manager, I want to review login history and available audit traces.	Medium
	US 38	**Team leave calendar view**	As a team leader, I want to view the leave calendar within my team scope so that I can plan availability.	Medium
	US 39	**Leave calendar filters**	As a team leader or HR manager, I want to filter leave information by employee, status, and date so that calendar views stay focused.	Low
	US 40	**HR leave calendar view**	As an HR manager, I want broader leave visibility across teams and employees so that I can supervise planning.	Medium
	US 41	**Request traceability**	As an HR manager, I want to trace request history so that each workflow decision remains auditable.	Medium

Table 1: Product Backlog (continued)

Sprint	Num	User Story	Description	Complexity
Sprint 5 Projects and Tasks	**US 42**	**Project creation**	As a team leader, I want to create projects so that I can organize work around clear objectives.	Medium
	US 43	**Project list**	As a user, I want to view the project list so that I can access the projects relevant to me.	Low
	US 44	**Task creation**	As a team leader, I want to create tasks so that project work can be broken into manageable actions.	Medium
	US 45	**Task assignment**	As a team leader, I want to assign tasks to team members so that project work is distributed properly.	Medium
	US 46	**Task status update**	As an assignee, I want to update task status so that task progress remains visible to the team.	Low
	US 47	**Task tracking**	As a user, I want to track task status so that I can monitor progress toward completion.	Medium
	US 48	**Project/task dashboard**	As a team leader, I want a project and task dashboard so that I can monitor delivery.	Medium
	US 49	**Project and task calendar view**	As a user, I want to view project and task dates in a calendar so that planning information is easier to follow.	Medium
	US 50	**Project-based task filters**	As a user, I want to filter tasks by project so that I can focus on a specific work context.	Low
	US 51	**Assignee task assignment notification and email**	As an assignee, I want a notification and email when a task is assigned to me.	Low
	US 52	**Team leader task completion notification and email**	As a team leader, I want a notification and email when a task is completed.	Low

Table 1: Product Backlog (continued)

Sprint	Num	User Story	Description	Complexity
Sprint 6 AI Assistance & Decision Support	**US 53**	**AI request assistant**	As an employee, I want an AI request assistant so that I can get guided help while preparing a request.	Medium
	US 54	**AI leave drafting**	As an employee, I want AI assistance to draft my leave request so that I can prepare it faster before confirming submission.	Medium
	US 55	**AI document drafting**	As an employee, I want AI assistance to draft my document request so that I can submit a clearer request after confirmation.	Medium
	US 56	**AI loan drafting**	As an employee, I want AI assistance to draft my loan request so that I can structure my request before confirming it.	Medium
	US 57	**AI authorization drafting**	As an employee, I want AI assistance to draft my authorization request so that I can prepare the request before submitting it.	Medium
	US 58	**TL request summary**	As a team leader, I want AI to summarize requests so that I can review them faster before making a human decision.	Medium
	US 59	**TL workload hints**	As a team leader, I want AI workload and conflict hints so that I can detect scheduling risks before validating team requests.	Medium
	US 60	**TL reassignment suggestions**	As a team leader, I want AI suggestions for task reassignment so that I can balance workload more effectively.	Medium
	US 61	**HR recommendation support**	As an HR manager, I want AI recommendation support so that I can review requests with decision assistance.	High
	US 62	**Explanation and traceability**	As a team leader or HR manager, I want AI explanations and traceability so that every suggestion remains understandable and auditable.	High

Table 1: Product Backlog (continued)

VI. Release Organization

The six planned sprints are grouped into three major releases to ensure progressive and coherent delivery of the platform.

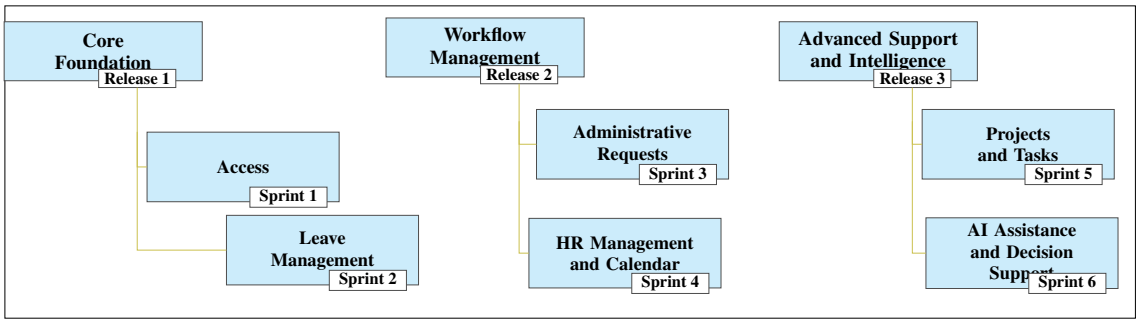


Figure 9: Organization of Releases

Conclusion

This chapter established the backlog-planning foundation of the project through actor identification, requirement structuring, user stories, and prioritized backlog definition. It provides the planning baseline for sprint execution and the progressive realization of the platform at ArabSoft.

Chapter 3

Release 1 – Foundations

Introduction

This chapter focuses on Release 1, which forms the functional and technical foundation of the ArabSoft internal platform. Through its use case diagrams, system sequence diagrams, state diagrams, class diagrams, and interface illustrations, this chapter presents how the first core functionalities of the platform are structured. It highlights the implementation logic of secure access management and the first complete administrative workflow. More specifically, Release 1 covers user registration, authentication, role assignment, account control, and leave request management. Together, these elements establish the first operational base of the platform for employees, team leaders, and HR managers.

I. Release Organization

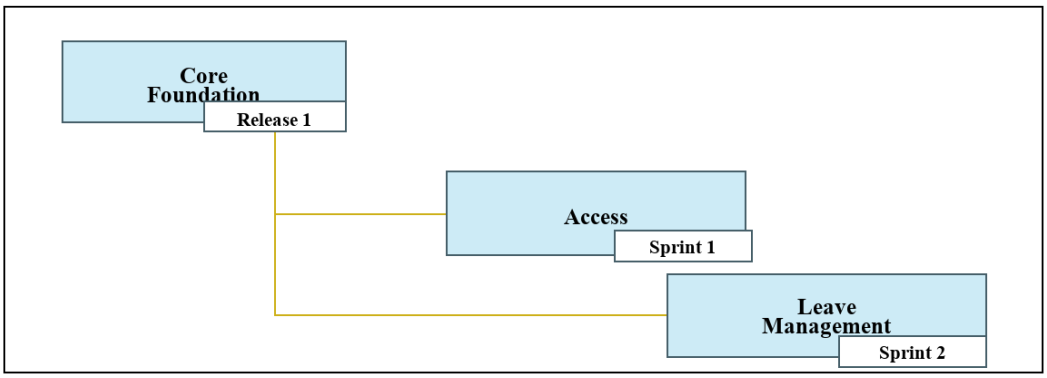


Figure 10: Release 1 Organization

II. Sprint 1: Access

1. Sprint Goal

This first sprint aims to establish the technical and functional access foundation of the platform. It implements the core access features: user registration, secure authentication, password reset, role assignment with notification and email, account activation or deactivation, profile update, and login history consultation. It also introduces the restricted **NEW_USER** state

to control onboarding before HR validation. The objective is to obtain an operational base for the next functional developments of the platform.

2. Visual Sprint Journal

The visual sprint journal shows the selected user stories of Sprint 1 and their progress in Jira.

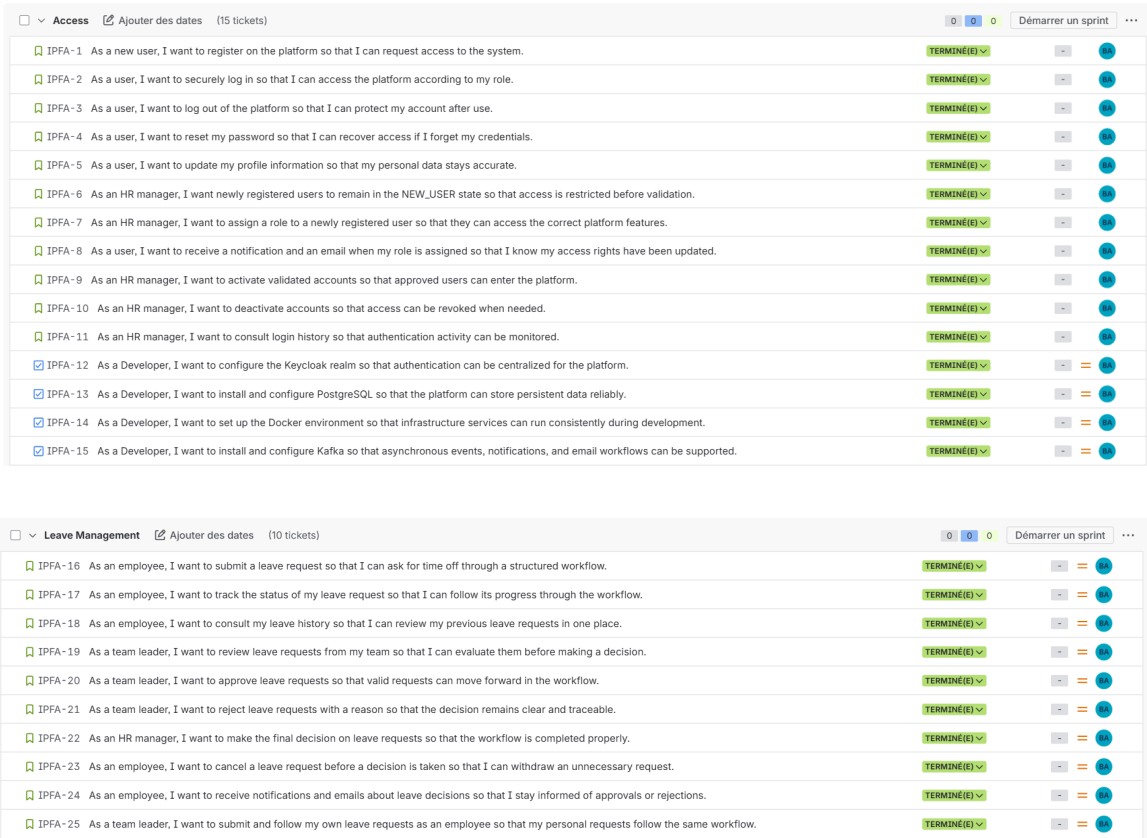


Figure 11: Jira Backlog – Sprint 1

3. Sprint Backlog

The table below presents the backlog of the first and second sprints.

Sprint	Stories	Tasks	Duration
Sprint 1	**Configure security and infrastructure**	Configure Keycloak realm, client, and roles	2d
		Set up PostgreSQL, Docker, and Kafka	
		Connect Spring Boot to PostgreSQL and Keycloak	
	Registration and access control	Implement user registration	3d
		Enforce the restricted **NEW_USER** state	
		Implement secure login	
		Implement frontend logout	
	Password recovery and profile management	Implement password reset	2d
		Implement profile update	
	HR account governance	Implement HR role assignment	2d
		Trigger role assignment notification and email	
		Implement account activation and deactivation	
	Access audit	Implement login history consultation	1d
Sprint 2	**Leave request workflow foundation**	Create the leave request entity, DTOs, repository, and service	3d
		Implement leave submission	
		Implement leave tracking	
		Implement leave history	
	Team leader review flow	Implement team review of leave requests	2d
		Implement leave approval	
		Implement leave rejection with reason	
	HR final decision flow	Implement HR final decision on leave requests	2d
		Handle workflow state transitions	
	Leave control and communication	Implement leave cancellation before decision	2d
		Implement leave decision notifications and emails	
	Team leader personal leave actions	Reuse employee-side leave actions for the team leader's own leave requests	1d

Table 7: Sprint Backlog – Release 1

4. Sprint Implementation

The implementation of Sprint 1 focused on building the secure access layer of the application. It introduced the mechanisms required to control how users enter the platform, how their identity is verified, and how their permissions are determined.

The first implemented functionality was **user registration**. This process allows a new

person to create an account in the platform by providing the required credentials and identification data. However, registration alone does not immediately grant operational access. After registration, the user is placed in a restricted **NEW_USER** state. This design ensures that no newly created account can access employee, team leader, or HR functionalities before validation by the HR manager.

The second key functionality was **secure login**. Authentication was designed to allow verified users to access the application through a controlled login process integrated with the security architecture of the system. Login does not simply check credentials; it also determines the current role associated with the account and applies the corresponding access rights in the frontend and backend. This guarantees that employees, team leaders, and HR managers each access only the parts of the platform that match their responsibilities.

The sprint also implemented a **frontend logout flow**. This action clears the local session state and stored authentication data on the client side. It contributes to account protection in the user interface, while the backend does not expose a dedicated logout endpoint.

Another important part of the sprint was the **password reset** mechanism. This feature allows users who can no longer access their account to recover it through a secure recovery process. From a usability perspective, this functionality reduces dependency on manual administrator intervention and improves continuity of access.

A central element of Sprint 1 is **HR role assignment**. Because newly registered users begin in a restricted state, HR intervention is necessary to transform a generic account into an operational one. The HR manager assigns the correct role to the user according to the person's function in the organization. This action determines whether the user becomes an employee, a team leader, or retains another platform role defined by the system. After the role is assigned, the concerned user is informed through a platform notification and an email message.

The sprint also covered **account activation and deactivation**. Activation allows HR to authorize access after validation, while deactivation allows HR to suspend access when necessary. This mechanism strengthens administrative control and contributes to security governance within the platform.

In addition, Sprint 1 implemented **profile update** and **login history consultation**. Profile update allows authenticated users to maintain their personal information, while login history

gives HR visibility over authentication activity for audit purposes.

From a modeling perspective, Sprint 1 is represented through several diagrams. The **use case diagram** presents the main interactions related to access management. The **system sequence diagrams** show the chronological flow of major scenarios such as registration, login, frontend logout, password reset, role assignment with user notification, and account activation or deactivation. The **state diagram** is particularly important in this sprint because it models the lifecycle of a user account, from registration to restricted state, active usage, and possible deactivation. Finally, the **domain model**, **participant class diagrams**, and **design class diagrams** help represent the structural and interaction logic of the access module.

Use Case Diagram – Sprint 1 Access

Figure 12: Use Case Diagram – Sprint 1 Access

System Sequence Diagram – User Registration

Figure 13: System Sequence Diagram – User Registration

System Sequence Diagram – Secure Login

Figure 14: System Sequence Diagram – Secure Login

System Sequence Diagram – Frontend Logout

Figure 15: System Sequence Diagram – Frontend Logout

System Sequence Diagram – Password Reset

Figure 16: System Sequence Diagram – Password Reset

System Sequence Diagram – HR Role Assignment

Figure 17: System Sequence Diagram – HR Role Assignment

System Sequence Diagram – Account Activation / Deactivation

Figure 18: System Sequence Diagram – Account Activation / Deactivation

State Diagram – User Account Lifecycle

Figure 19: State Diagram – User Account Lifecycle

Domain Model – Sprint 1

Figure 20: Domain Model – Sprint 1

Participant Class Diagram – Secure Login

Figure 21: Participant Class Diagram – Secure Login

Participant Class Diagram – HR Role Assignment

Figure 22: Participant Class Diagram – HR Role Assignment

Design Class Diagram – Secure Login

Figure 23: Design Class Diagram – Secure Login

Design Class Diagram – HR Role Assignment

Figure 24: Design Class Diagram – HR Role Assignment

5. Sprint Review

At the end of Sprint 1, the platform provided a first usable access foundation. The implemented functionalities made it possible to create accounts, restrict newly registered users until HR validation, authenticate authorized users securely, recover passwords, update profiles, assign roles with notification and email, consult login history, and manage account status. This result was essential because it established the operational entry point of the system and prepared the environment for business workflows in the following sprint.

The sprint therefore met its main objective by delivering a controlled and role-aware access mechanism aligned with the needs of the platform.

6. Sprint Retrospective

Sprint 1 confirmed the importance of securing the application foundation before introducing business features. It also highlighted that user lifecycle management is not limited to

login and frontend session clearing, but includes onboarding control, role assignment communication, administrative activation logic, profile maintenance, and audit visibility. One important lesson from this sprint is that early clarification of user states and permissions reduces complexity in later modules and helps maintain consistency between frontend behavior and backend authorization rules.

III. Sprint 2: Leave Management

1. Sprint Goal

This sprint aims to implement the first complete administrative workflow of the platform through leave management. It allows the employee to submit and track leave requests, while enabling the team leader to review them and HR to make the final decision. The sprint also structures request states, cancellation before decision, approval levels, decision notifications, and traceability within a centralized workflow. Through its diagrams and interface elements, it shows how leave management is modeled and integrated into the system. It therefore represents the first concrete business process delivered by the platform.

2. Visual Sprint Journal

The visual sprint journal shows the selected user stories of Sprint 2 and their progress in Jira.

Jira Backlog – Sprint 2

Figure 25: Jira Backlog – Sprint 2

3. Sprint Implementation

The implementation of Sprint 2 focused on transforming leave management into a structured digital workflow integrated into the platform. This sprint introduced the first full approval chain involving the employee, the team leader, and the HR manager.

The first step of this workflow is **leave request submission**. The employee can create a request by specifying the leave information required by the platform. This action transforms what is traditionally managed through fragmented communication into a centralized and traceable process. After submission, the employee can also cancel the request before a decision is taken, which keeps the workflow flexible without bypassing the approval rules.

Once submitted, the leave request enters the review workflow and becomes visible to the appropriate decision makers. The employee can also **track the request status** and consult the leave history, which improves transparency and reduces uncertainty. Instead of relying on informal follow-up, the user can directly consult whether the request is pending, approved, rejected, or cancelled.

A key part of Sprint 2 is the **team leader review** stage. At this point, the team leader examines leave requests submitted by team members and evaluates them according to operational constraints such as team availability and workload continuity. The team leader can then **approve or reject** the request. When a rejection occurs, a reason is required, which keeps the decision explicit and understandable and supports traceability and communication clarity.

After the managerial stage, the leave request reaches the **HR final decision** stage. The HR manager performs the final validation of the request in order to complete the official workflow. This second approval level is important because it aligns the process with organizational control requirements and formalizes the final outcome of the request.

Sprint 2 also includes **employee notification and email communication**. When a leave request is approved or rejected, the employee receives an in-app notification through the platform notification system. The system also sends an email to the employee for leave approval or rejection.

The sprint also respects the fact that the team leader is a platform user with an employee profile. When acting as a requester for personal leave, the team leader follows the employee-side logic for submission, tracking, history consultation, cancellation before decision, and decision communication.

An essential contribution of Sprint 2 is the explicit modeling of the **leave request lifecycle**. Through the state diagram, the workflow can be represented from submission to review, employee cancellation, final approval, or rejection. This state-based representation helps struc-

ture both the business logic and the understanding of the process in the report.

The sprint is documented through several diagram families. The **use case diagram** presents the main leave-related interactions. The **system sequence diagrams** describe how requests move through the platform during submission, cancellation, review, approval, rejection, notification, and final validation. The **state diagram** shows the lifecycle of the leave request. The **domain model**, **participant class diagrams**, and **design class diagrams** provide the structural representation of the leave module and its approval workflow.

Through this sprint, the platform begins to demonstrate its real business value. It no longer acts only as an access-controlled system, but as an operational tool capable of structuring internal administrative workflows and improving coordination between stakeholders.

Use Case Diagram – Sprint 2 Leave Management

Figure 26: Use Case Diagram – Sprint 2 Leave Management

System Sequence Diagram – Submit Leave Request

Figure 27: System Sequence Diagram – Submit Leave Request

System Sequence Diagram – Cancel Leave Request

Figure 28: System Sequence Diagram – Cancel Leave Request

System Sequence Diagram – Track Leave Request Status

Figure 29: System Sequence Diagram – Track Leave Request Status

System Sequence Diagram – Team Leader Reviews Leave Request

Figure 30: System Sequence Diagram – Team Leader Reviews Leave Request

System Sequence Diagram – Team Leader Approves / Rejects Leave Request

Figure 31: System Sequence Diagram – Team Leader Approves / Rejects Leave Request

System Sequence Diagram – HR Final Decision on Leave Request

Figure 32: System Sequence Diagram – HR Final Decision on Leave Request

State Diagram – Leave Request Lifecycle

Figure 33: State Diagram – Leave Request Lifecycle

Domain Model – Sprint 2

Figure 34: Domain Model – Sprint 2

Participant Class Diagram – Submit Leave Request

Figure 35: Participant Class Diagram – Submit Leave Request

Participant Class Diagram – Team Review of Leave Request

Figure 36: Participant Class Diagram – Team Review of Leave Request

Participant Class Diagram – HR Final Decision

Figure 37: Participant Class Diagram – HR Final Decision

Design Class Diagram – Submit Leave Request

Figure 38: Design Class Diagram – Submit Leave Request

Design Class Diagram – Leave Approval Workflow

Figure 39: Design Class Diagram – Leave Approval Workflow

4. Sprint Review

At the end of Sprint 2, the platform provided a first complete administrative workflow. Employees could submit, cancel before decision, track, and consult the history of leave requests, team leaders could review and decide at the managerial level with a required reason

for rejection, and HR managers could finalize the process. Team leaders could also use the employee-side leave actions for their own personal requests. Employees were informed of approval or rejection through in-app notifications and email messages. This result demonstrated the ability of the platform to support a multi-level approval mechanism while maintaining clarity, status visibility, and decision traceability.

Sprint 2 therefore achieved its main objective by delivering the first concrete business process of the application.

5. Sprint Retrospective

Sprint 2 showed that workflow implementation requires more than simple form handling. It involves actor coordination, lifecycle control, cancellation rules, status consistency, and clear separation of responsibilities between review levels. One major lesson from this sprint is that modeling request states and validation stages early helps reduce ambiguity in both implementation and documentation. It also reinforces the importance of traceability in administrative processes, especially when multiple actors intervene in the same decision chain.

Conclusion

This chapter presented the realization of **Release 1**, which established the first operational foundation of the ArabSoft internal platform. Through Sprint 1, the project introduced secure access management, controlled onboarding, role assignment with notification and email, account state supervision, profile maintenance, and login audit visibility. Through Sprint 2, it delivered the first complete business workflow by implementing structured leave management with multi-level validation, cancellation before decision, team leader personal leave actions, request traceability, in-app employee notifications, and email communication for leave approval or rejection.

The results of this release are essential for the remainder of the project. They provide both the technical base required for future modules and the first concrete demonstration of how the platform can improve internal administrative management. The next release will build on this foundation by extending the platform toward additional request types and broader functional coverage.